212791: monzogranite gneiss, New Holland mine

(Yilgarn Craton granites, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Sir Samuel (SG 51-13), Depot Springs (2942)
MGA Zone 51, 250125E 6901420N

Sampled on 24 May 2014

This sample was collected from low outcrop on Pinnacles Station, about 6.0 km north of Five Mile Well, 3.1 km west of the New Holland mine openpits, and 2.5 km due south of the Sandstone–Agnew Road.

Geological context

The unit sampled is a granitic gneiss (Fig. 1) in the hanging wall of the c. 2660 Ma Waroonga Shear Zone (WSZ), along the eastern margin of the Youanmi Terrane (Zibra et al., 2017). The WSZ is a west-dipping, low-angle, listric structure that postdates regional east-dipping fabrics (Zibra et al., 2017). Shearing postdates the <2662 Ma Scotty Creek Formation in the footwall of the WSZ (GA 2001967052C, Dunphy et al., 2003), and predates emplacement of the non-deformed 2655 ± 4 Ma Breakaway Well Granodiorite (GA 95969764B, Fletcher et al., 2001). A syntectonic syenogranite sampled about 36 km to the north is deformed along the WSZ and has an igneous crystallization age of 2661 ± 6 Ma (GSWA 209610, Lu et al., 2016; Zibra et al., 2017). The present sample was collected from a syntectonic pluton, which occupies a similar structural position to GSWA 209610 in the hanging wall of the WSZ and exhibits a tectonic fabric that developed during melt-present thrusting (Zibra et al., 2017).

Petrographic description

The sample is a foliated, medium-grained monzogranite gneiss, consisting of about 40% quartz, 30% plagioclase, 25% K-feldspar, 2% biotite, 1–2% muscovite, 1% magnetite–hematite, and trace zircon. Quartz forms elongate to irregular ribbons up to 8 mm long that include anhedral, moderately strained grains up to 1.5 mm in size, and also grains up to 1 mm in size that exhibit mutual, low-energy curved boundaries. Plagioclase (andesine, $An_{30–34}$) is anhedral, equant to slightly tabular, up to 2 mm in size, and exhibits deformation twins in some grains. K-feldspar is anhedral, equant to slightly elongate, up to 3 mm in size, and finely microperthitic; many of the larger grains display fine microcline twinning. Both feldspars also occur as much smaller (mostly <0.3 mm in diameter), polygonal grains with quartz in irregular, elongate ribbons up to 5 mm long. Orange–brown biotite forms subhedral laths up to 0.5 mm long that are strongly aligned in narrow folia. Muscovite also occurs as subhedral laths to 0.5 mm long, some aligned in the biotite folia and others along feldspar grain boundaries. Magnetite is subhedral, up to 1 mm diameter, and mostly replaced by hematite. The sample is not altered apart from minor sericite alteration of plagioclase. The rock has been strongly deformed to produce the foliation defined by elongate quartz and biotite folia, and widespread feldspar recrystallization. Deformation and partial recovery, indicated by the aggregates of small polygonal grains, are inferred to have occurred during cooling from high-temperature conditions.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown or opaque, anhedral to subhedral, and variably rounded. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, most crystals are dominated by high-U, metamict domains, and some exhibit alteration to luminescent zircon. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 20–21 and 26 September 2019, using SHRIMP-B. Analyses 1.1 to 11.1 (spot numbers 1–11) were obtained during the first session, together with six analyses of the M257 standard, of which five analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.59% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.35% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.71%. Analyses 12.1 to 5.2 (spot numbers 12–15) were obtained during the second session, together with four analyses of the M257 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.67% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Figure 1. Outcrop image of sample 212791: monzogranite gneiss, New Holland mine

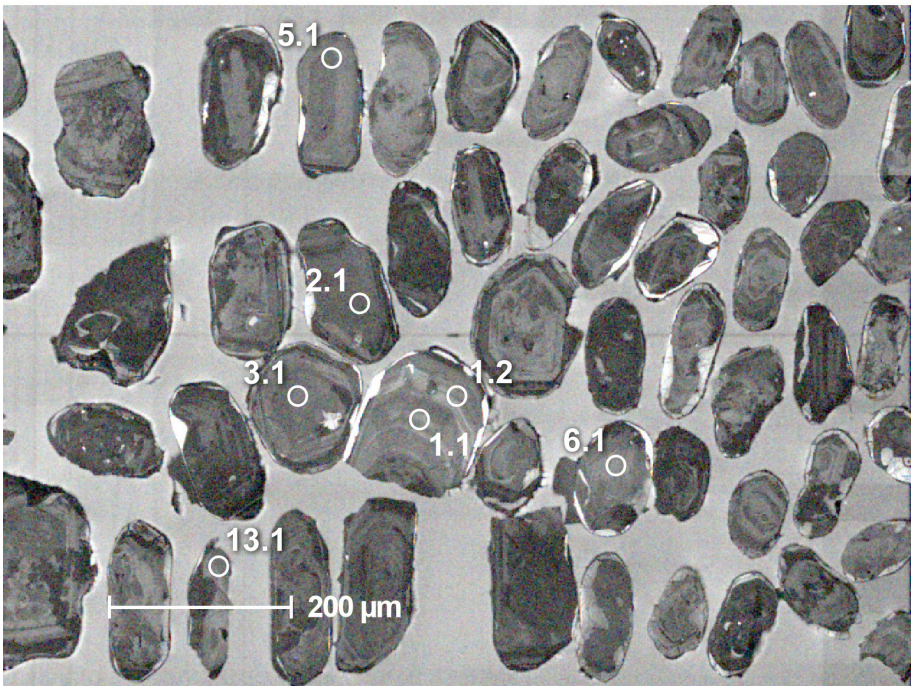

Figure 2. Cathodoluminescence image of representative zircons from sample 212791: monzogranite gneiss, New Holland mine. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 212791: monzogranite gneiss, New Holland mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
X	13	1.2	118	48	0.41	0.154	1.925	0.028	0.19597	0.00144	1.928	0.029	0.19460	0.00155	2694	34	2781	13	3.1
X	1	1.1	120	48	0.40	0.068	1.881	0.020	0.19689	0.00096	1.882	0.020	0.19629	0.00099	2747	25	2796	8	1.7
X	5	5.1	137	75	0.55	0.116	1.913	0.020	0.19768	0.00086	1.916	0.020	0.19666	0.00091	2708	23	2799	8	3.2
X	8	8.1	347	195	0.56	0.064	1.911	0.016	0.19807	0.00056	1.912	0.016	0.19751	0.00058	2712	20	2806	5	3.4
X	6	6.1	218	98	0.45	0.094	1.933	0.018	0.19855	0.00070	1.935	0.018	0.19772	0.00073	2685	21	2807	6	4.3
X	7	7.1	1935	951	0.49	0.003	1.791	0.013	0.19918	0.00024	1.791	0.013	0.19915	0.00024	2860	18	2819	2	-1.4
X	2	2.1	255	129	0.51	0.174	1.903	0.017	0.20070	0.00065	1.906	0.017	0.19916	0.00072	2719	21	2819	6	3.6
X	11	11.1	732	361	0.49	0.003	1.877	0.015	0.19941	0.00039	1.877	0.015	0.19938	0.00039	2753	18	2821	3	2.4
X	9	9.1	588	285	0.48	0.041	1.842	0.015	0.20009	0.00044	1.842	0.015	0.19973	0.00045	2795	19	2824	4	1.0
X	3	3.1	673	508	0.75	0.035	1.794	0.014	0.20252	0.00039	1.795	0.014	0.20221	0.00040	2855	19	2844	3	-0.4
X	15	5.2	99	38	0.38	0.000	1.912	0.030	0.20280	0.00154	1.912	0.030	0.20280	0.00154	2712	36	2849	12	4.8
D	14	13.1	175	66	0.38	0.460	2.311	0.030	0.20205	0.00133	2.321	0.030	0.19795	0.00160	2309	27	2809	13	17.8
D	10	10.1	143	91	0.63	0.204	2.131	0.022	0.19985	0.00093	2.135	0.022	0.19805	0.00101	2476	22	2810	8	11.9
D	12	12.1	149	160	1.07	0.314	2.670	0.056	0.20467	0.00157	2.678	0.056	0.20187	0.00179	2045	37	2841	14	28.0
D	4	4.1	812	589	0.73	0.013	3.519	0.032	0.13140	0.00038	3.519	0.032	0.13128	0.00039	1612	13	2115	5	23.8

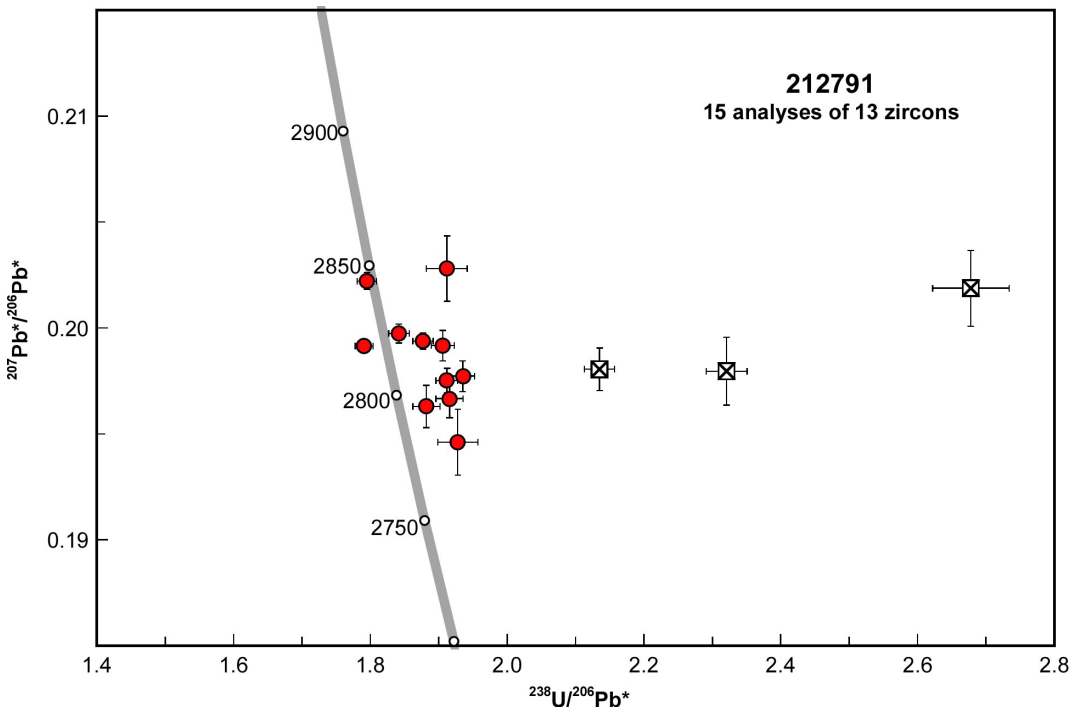

Figure 3. U–Pb analytical data for sample 212791: monzogranite gneiss, New Holland mine. Red circles indicate Group X (xenocrystic zircons); crossed squares indicate Group D (discordance >5%). One strongly discordant analysis in Group D is not shown

Results

Fifteen analyses were obtained from 13 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Four analyses are >5% discordant. The dates obtained from these four analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 11 analyses form a single group, based on their ^{207}Pb*/^{206}Pb* ratios.

Group X comprises 11 analyses of 10 zircons (Table 1), which yield ^{207}Pb*/^{206}Pb* dates of 2849–2781 Ma. Four analyses of four zircons yield a weighted mean ^{207}Pb*/^{206}Pb* date of 2820 ± 3 Ma (MSWD = 0.47), and five analyses of four zircons yield a weighted mean ^{207}Pb*/^{206}Pb* date of 2802 ± 6 Ma (MSWD = 1.2).

The dates of 2849–2781 Ma for the 11 analyses in Group X are older than the c. 2660 Ma age estimated for this syntectonic pluton based on observations of melt-present shearing (Zibra et al., 2017), hence are interpreted as the ages of inherited zircons. The dates of 2820 ± 3 and 2802 ± 6 Ma are interpreted as the possible ages of inherited components.

References

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Recommended reference for this publication

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